

Chapter 8

Worksheet

Instructions:

Please complete all activities in this worksheet. Write your answers in the spaces provided or on separate sheets if needed.

Worksheet for Chapter 8: Building Simple Apps and Games

This worksheet is designed to reinforce the concepts covered in Chapter 8 through a mix of interactive activities, short-answer questions, hands-on tasks, and creative challenges. Complete each section to test your understanding and apply your skills in app and game development.

Part 1: Concept Matching Activity

Match the key terms or tools from Chapter 8 with their correct descriptions. Draw a line from each term on the left to the corresponding description on the right. (1 point each)

Term/Tool	Description
• 1. User Interface (UI)	• A) A Python library for creating 2D games with graphics and sound
• 2. Pygame	• B) The process of finding and fixing errors in code
• 3. MIT App Inventor	• C) The visual elements of an app, like buttons and text, that users interact with
• 4. Debugging	• D) A block-based tool for building mobile apps without typing code
• 5. User Experience (UX)	• E) How a user feels while using an app, focusing on ease and enjoyment
• 6. Testing	• F) Trying out an app or game to ensure it works and identify issues

Term/Tool

Description

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- D) A block-based tool for building mobile apps without typing code
- 5. User Experience (UX)
- E) How a user feels while using an app, focusing on ease and enjoyment
- 6. Testing
- F) Trying out an app or game to ensure it works and identify issues

Part 2: Short-Answer Questions

Answer the following questions in 2-3 sentences. Be clear and concise. (2 points each)

- **Explain the difference between User Interface (UI) and User Experience (UX) in app development.**
- **Why is Pygame a good choice for beginners learning game development?**
- **Describe one way you could improve the “Dodge the Squares!” game based on user feedback.**

Part 3: Hands-On Coding Activity

Apply what you’ve learned by completing a small coding task. Choose **one** of the following activities based on your interest in Pygame or MIT App Inventor. Write or describe your steps and results. (5 points)

Option 1: Pygame Enhancement for “Dodge the Squares!” Add a simple score counter to the “Dodge the Squares!” game that increases by 1 each time an enemy square reaches the bottom of the screen. Write the additional code snippet you would add to the game loop (or describe where and how you would add it if you can’t code right now). Test it mentally or on a computer if possible, and note any challenges you might face. **Your Code/Description:**

Challenges or Observations:

Option 2: MIT App Inventor Feature for “Random Meme Picker” Add a new feature to the “Random Meme Picker” app, such as a second button that resets the app to its initial state (e.g., clears the quote or sets a default message). Describe the steps to add this in the Designer and Blocks mode (or sketch the blocks if you can’t access the tool). Note how this improves the user experience. **Your Steps/Description:**

Improvement to User Experience:

Part 4: Creative Challenge

Let your imagination run wild! Complete this activity to think like a developer. (3 points)

Dream App or Game Idea: Imagine you could build any app or game in the world. Write a short description (3-5 sentences) of your idea, including its purpose, target audience (e.g., teens, kids, etc.), and one key feature that makes it unique. Save this as dream_project.txt if working digitally, or write it below. **Your Idea:**

Part 5: Reflection

Reflect on your learning from Chapter 8. Answer in 3-5 sentences. (2 points)

What was the most interesting or challenging part of this chapter for you, and why? How do you think building apps or games could help you in the future (e.g., hobbies, career, or problem-solving)?